ZAKI HATTA

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Education

Padang State University (UNP)
Bachelor Degree in Informatics Engineering, 3.60/4.00

- Bachelor's Thesis : Design and Development of a Multi-Platform Tsunami Early Warning and Evacuation System Using Ray Casting and Dijkstra's Algorithm in Padang City (2026)
- Industrial Field Experience (PLI) at PT. Mitra Tech Indonesia (2025)

Work Experience

PT Mitra Tech Indonesia Aug 2025 – Oct 2025
WebGIS Intern

- Processed and cleaned spatial data from SHP, CAD, and Excel formats using ArcGIS Pro, including geometry correction, coordinate system standarization, and on-screen digitization of plantation blocks and infrastructure.
- Contributed to the development a WebGIS-based spatial asset management system for the plantation industry, collaborating with GIS Engineers and Web Developers.

Selected Projects

AssistCare Indonesia Website Apr 2023 – Jun 2023
Fullstack Developer

- Pioneered a digital revolution at Assist Care Indonesia, leading the architectural design and development of a cutting-edge Next.js based web platform; achieved a remarkable 40% improvement in operational performance and doubled processing speed for international medical services.

Seven Star Production Website Aug 2023 – Sep 2023
Web Developer

- Participate in developing the appearance of the Seven Star Production website using WordPress, making the website look neater and more attractive so that it is easier for users to use.

YPRO Jun 2025 – Jul 2025
Web Developer

- Developed a corporate website for YPRO, a bag manufacturing and promotional merchandise company, using WordPress and Divi Builder.
- Collaborated directly with the client to align website design and content with business requirements.

Seven Star Production Ticketing Platform Sept 2026 – Oct 2026
Frontend Developer

- Engineered a highly responsive online ticketing frontend for Sevenstarproduction using TypeScript, Next.js, and Tailwind CSS, delivering a minimalist and modern user interface that streamlined the ticketing purchasing process

Padang Rescue Application Jan 2026 – Sep 2026

Software Engineer

- Engineered a real-time tsunami mitigation mobile GIS system using React Natice and spatial mapping tools, enhancing emergency preparedness and geospasial data processing reliability.
- Optimized real-time hazard warning distribution by utilizing persistent WebSocket connections, achieving an 18x speed increase and reducing network latency from 677.57 ms to just 37.39 ms.
- Modified the evacuation routing system using the Dijkstra algorithm enhanced with a Failover Node Elimination (Top-K) mechanism, successfully rerouting users to alternative shelters in 0.4066 ms to prevent mass bottlenecks.

Additional

- **Certifications:** Mikrotik Certified Network Associate (2024), HCIA-AI V3.5 Course – Huawei Talent Online (2024)**;**
- **Languages:** English (Professional Proficiency), Indonesian (Native).
- **Hard Skills:** Cloud Engineering & DevOps: AWS, Docker; Experienced Programming Languages: JavaScript, TypeScript, Python, Java, C, C++; Web Development: React, Node.js; Database Management: SQL & No-SQL Databases; Version Control and Collaboration Tools: GIT.
- **Soft Skills:** Teamwork, Leadership, Team and Timeline Management, Verbal and Written Communication, and Collaboration.
- **Research Paper:** Rancang Bangun Sistem Peringatan Dini dan Evakuasi Tsunami Multi-Platform Menggunakan Algoritma Ray Casting dan Dijkstra di Kota Padang (2026)